

Sulagna Ghosh

Department of Statistics, The University of Chicago
sulagnag@uchicago.edu · sulagna-ghosh.github.io · Google Scholar

Bayesian machine learning · temporal point processes · scalable variational inference · customer behavior analysis

Education

- 2022 – present **Ph.D. in Statistics**, The University of Chicago
Advisor: Aaron Schein. Expected graduation: June 2027.
Dissertation: Scalable gradient-based variational inference for flexible point process modeling of spatiotemporal data.
- 2018 – 2020 **M.Stat.** (Applied Statistics, Finance), Indian Statistical Institute, Kolkata
Thesis: *A study on retrospective inference from recalled time-to-event data*.
Advisor: Debasis Sengupta. Overall: 91.2%.
- 2015 – 2018 **B.Sc. in Statistics**, Presidency University, Kolkata. CGPA 9.81/10.

Publications and Preprints

Under review / revision

- **S. Ghosh** and A. Schein. Scalable amortized variational inference for non-Poisson buy-’til-you-die models. Under review at *International Journal of Research in Marketing*, 2026. arXiv:2608.19022.
- **S. Ghosh**, N. Ignatiadis, F. Koehler, and A. Lee. Stein’s unbiased risk estimate and Hyvärinen’s score matching. Under revision at *Journal of the Royal Statistical Society: Series B*, 2025. arXiv:2502.20123.

Published

- M. Biswas, C. R. Bhat, **S. Ghosh**, and A. R. Pinjari. Choice models with stochastic variables and random coefficients. *Journal of Choice Modelling*, 51, 2024. doi:10.1016/j.jocm.2024.100488.

In preparation

- **S. Ghosh** and A. Schein. Flexible and scalable Bayesian modeling of spatiotemporal event data with low-rank Neyman–Scott tensor processes.

Research Experience

- 2023 – present **Doctoral research**, The University of Chicago. Advisor: Aaron Schein.
Scalable Bayesian models for large-scale marked event data from international relations, combining Neyman–Scott point processes, non-negative tensor decomposition, and gradient-based variational inference to recover latent “major stories”. Amortized variational inference for non-Poisson buy-’til-you-die models of customer behavior.
- 2024 – 2025 **Score matching Empirical Bayes**, with Nikos Ignatiadis, The University of Chicago.
Showed that SURE-trained denoisers, equivalent to score matching, attain near-parametric regret rates in empirical Bayes and outperform the NPMLE under misspecification.
- 2019 – 2020 **Retrospective inference from recalled time-to-event data**, with Debasis Sengupta, Indian Statistical Institute. Master’s thesis.
A model that accounts for both inaccuracy and imprecision in recall while addressing confounding between correctly and incorrectly recalled observations; applied to survey data on age at menarche.
- Summer 2019 **Project trainee**, Indian Institute of Science (IISc), Bengaluru. Advisor: A. R. Pinjari.
Contributed to a discrete choice model framework that jointly accommodates stochastic alternative attributes and random coefficients, showing via simulation and a truck route-choice application that omitting either biases parameter estimates.

Presentations

- **Poster.** Flexible and scalable Bayesian modeling of spatiotemporal event data with low-rank Neyman–Scott tensor processes. ISBA 2026 World Meeting, Nagoya, Japan, 2026. **Best Poster Prize.**
- **Talk.** Scalable amortized variational inference for non-Poisson buy-’til-you-die models. Mini-Seminar, Department of Statistics, The University of Chicago, 2024.

Honors and Awards

2026	Best Poster Prize, ISBA 2026 World Meeting, Nagoya, Japan.
2020	Shortlisted, P. C. Mahalanobis Memorial Gold Medal (outstanding performance in M.Stat.), ISI Kolkata.
2019, 2020	Teachers’ Committee Award for Outstanding Academic Performance, ISI Kolkata.
2018	All India Rank 1, M.Stat. entrance examination, Indian Statistical Institute.
2018	All India Rank 4, IIT-JAM examination in Mathematical Statistics.
2018	Dr. Syama Prasad Mookherjee Gold Medal, Presidency Alumni Association Gold Medal, and departmental Gold Medal for the highest B.Sc. record, Presidency University.
2015 – 2018	INSPIRE-SHE Scholarship (for top 1% in 12th board examinations), Department of Science and Technology, Government of India.

Teaching

- Teaching assistant**, Department of Statistics, The University of Chicago, 2023 – present.
- STAT 34300: Applied Linear Statistical Methods (graduate), Nikos Ignatiadis. Autumn 2024–2026.
 - STAT 22200: Linear Models and Experimental Design (undergraduate), Fei Liu. Spring 2025–2026.
 - STAT 34700: Generalized Linear Models (graduate), Jingshu Wang. Winter 2024.
 - STAT 22000: Statistical Methods and Applications (undergraduate), Fei Liu and Kendra Burbank. Spring and Autumn 2023.
 - STAT 22600: Analysis of Categorical Data (undergraduate), Yibi Huang. Winter 2023.

Professional Experience

2020 – 2022	Risk Modeling and Analytics Specialist , UBS Business Solutions, Mumbai. AI, Monitoring & Surveillance team. Validated statistical and machine-learning models for transaction monitoring and detection of suspicious activity.
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Skills

Programming	Python (PyTorch, NumPy), R, C.
Methods	Variational inference, point processes, tensor decomposition, empirical Bayes.
Languages	English, Bengali, Hindi (full professional proficiency).

Service and Activities

- Member, International Society for Bayesian Analysis (ISBA), 2026–present.
- Co-president (2025–26) and decoration lead (2024–25), Kheyal, the Bengali student organization at The University of Chicago.
- Mentor for new analysts, UBS Business Solutions, Mumbai, 2021–22.